Chapter 5 - System Implementation

5.1 Introduction

The implementation phase involved developing the AI-Powered Customer Support & Service Automation application using the Salesforce Lightning Platform. The application was implemented by creating custom objects, configuring fields and relationships, applying validation rules, developing automation using Salesforce Flows, configuring queues for case assignment, and integrating Artificial Intelligence through Salesforce Prompt Builder.

The implementation followed a modular approach, ensuring that each component was independently configured and thoroughly tested before integrating it into the overall system.

5.2 Salesforce Environment Setup

The development environment was established using Salesforce Developer Edition, providing access to Salesforce Lightning Experience and AI capabilities. The Salesforce platform was configured to support custom object development, automation, AI integration, and security management.

The following Salesforce tools were used:

Tool	Purpose
Salesforce Developer Edition	Development Platform
Object Manager	Custom Object Configuration
Lightning App Builder	User Interface Development
Flow Builder	Business Process Automation
Prompt Builder	AI Summary Generation
Queues	Automatic Case Assignment
Validation Rules	Data Validation
Reports & Dashboards	Data Analysis

5.3 Custom Object Implementation

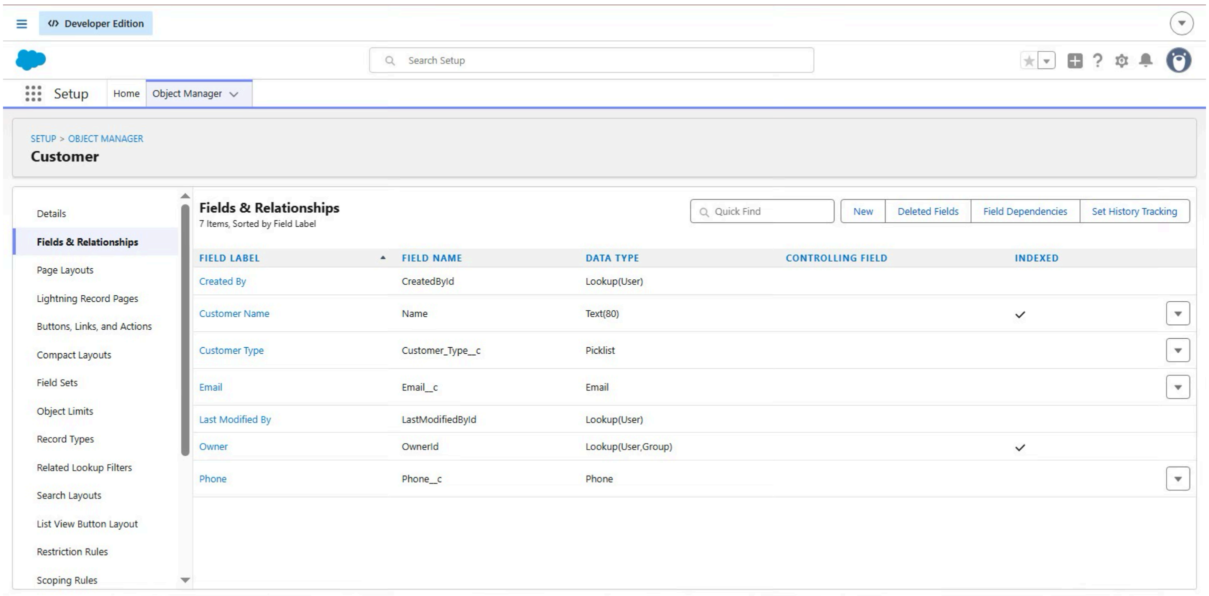
Four custom objects were created to represent the core entities of the application.

Customer

The Customer object stores customer information including contact details and customer type.

Fields implemented:

- Customer Name
- Email
- Phone
- Customer Type

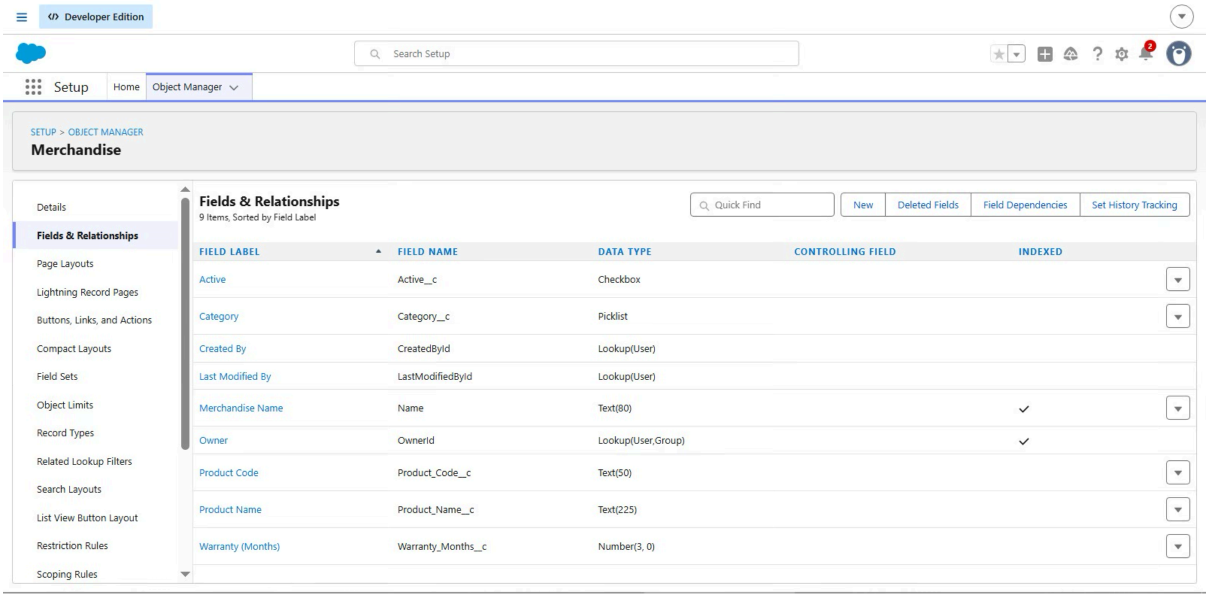


Product

The Product object stores product information including warranty and category details.

Fields implemented:

- Product Name
- Product Code
- Category
- Warranty (Months)
- Active

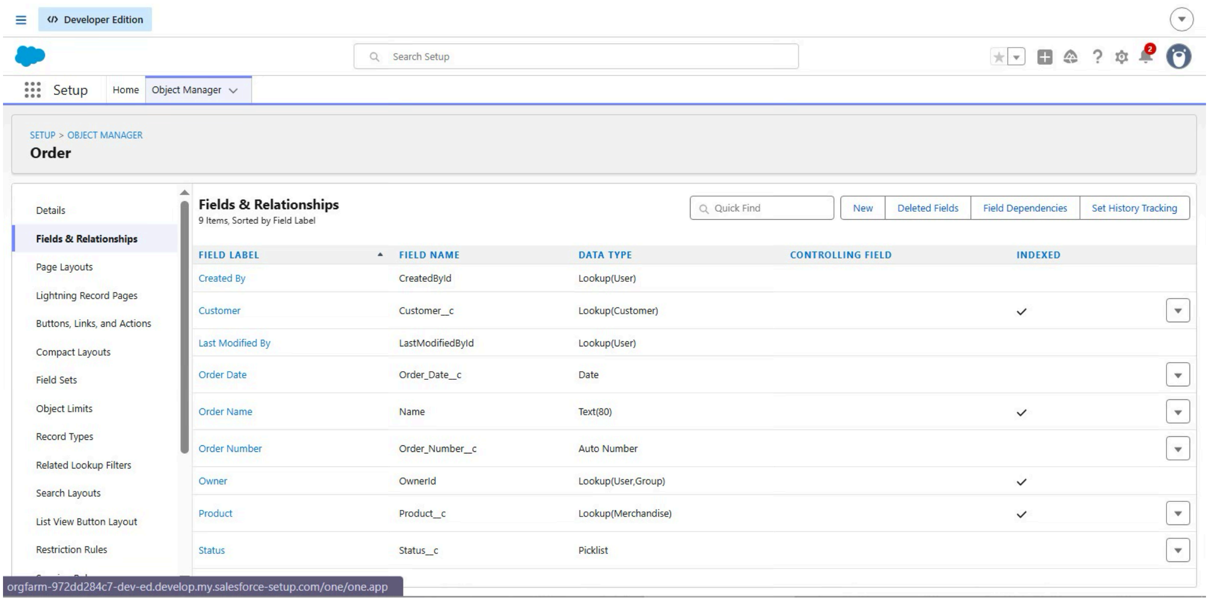


Order

The Order object stores customer purchase information and establishes relationships between customers and products.

Fields implemented:

- Order Number
- Order Date
- Status
- Customer Lookup
- Product Lookup

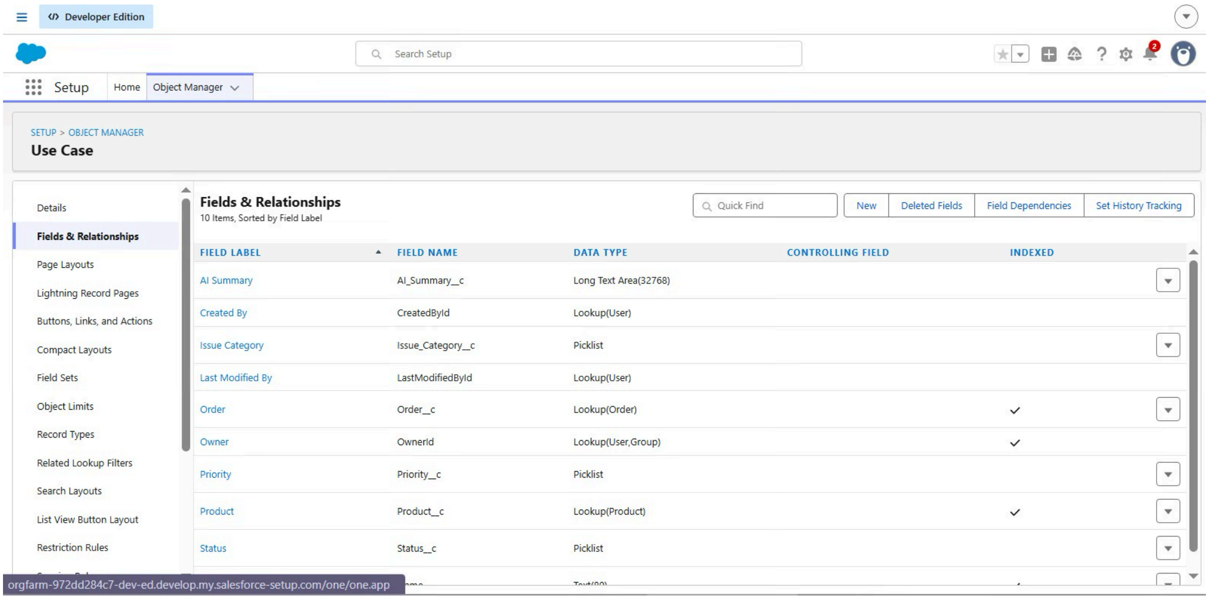


Use Case

The Use Case object stores customer support requests and AI-generated summaries.

Fields implemented:

- Product Lookup
- Order Lookup
- Priority
- Issue Category
- AI Summary



5.4 Field Configuration

Custom fields were configured using Salesforce Object Manager. Appropriate data types such as Text, Email, Phone, Number, Checkbox, Picklist, Lookup Relationship, Date, Auto Number, and Long Text Area were selected according to business requirements.

Lookup relationships were established between Customer, Product, Order, and Use Case objects to maintain data consistency and support relational data management.

5.5 Validation Rule Implementation

Validation Rules were implemented to ensure accurate and consistent data entry.

Damage_Requires_Priority

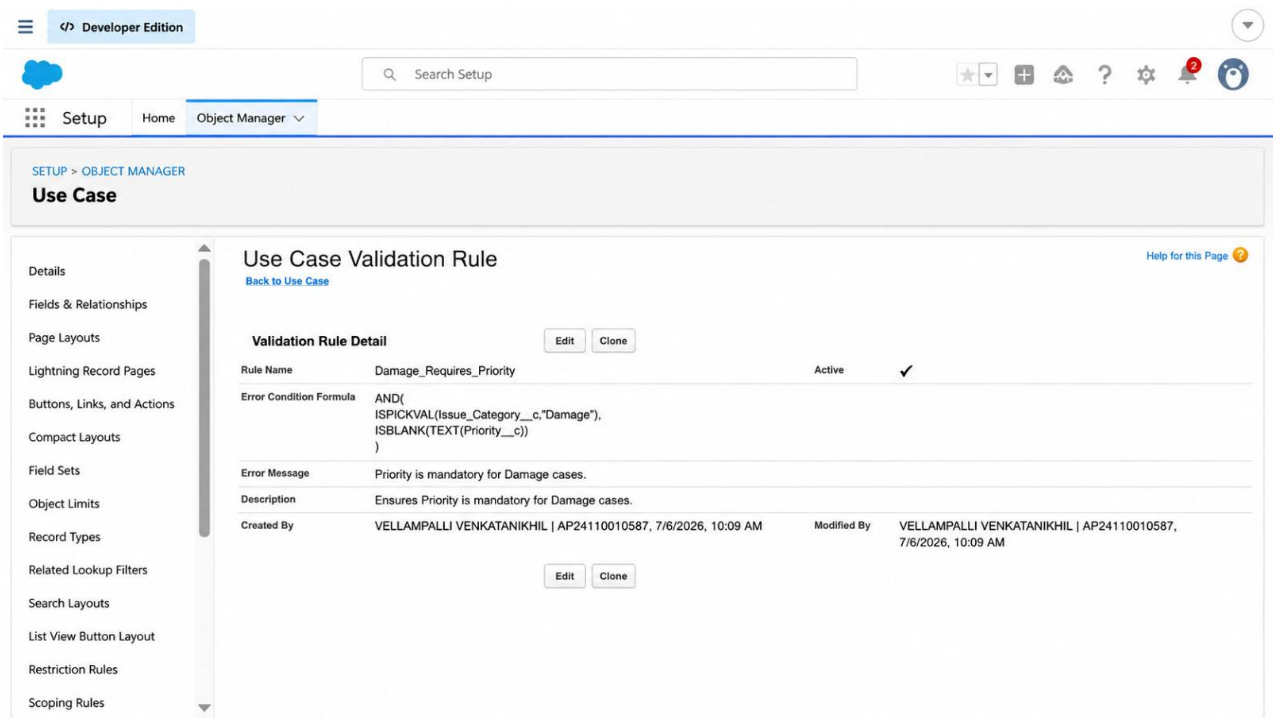
This validation rule prevents users from creating Damage cases without selecting a Priority.

Business Rule: Damage cases must always have a priority assigned.

Warranty_Delivery_Requires_Priority

This validation rule ensures that Warranty and Delivery cases also require a Priority before the record can be saved.

Business Rule: Warranty and Delivery cases cannot be created without specifying Priority.



5.6 Flow Implementation

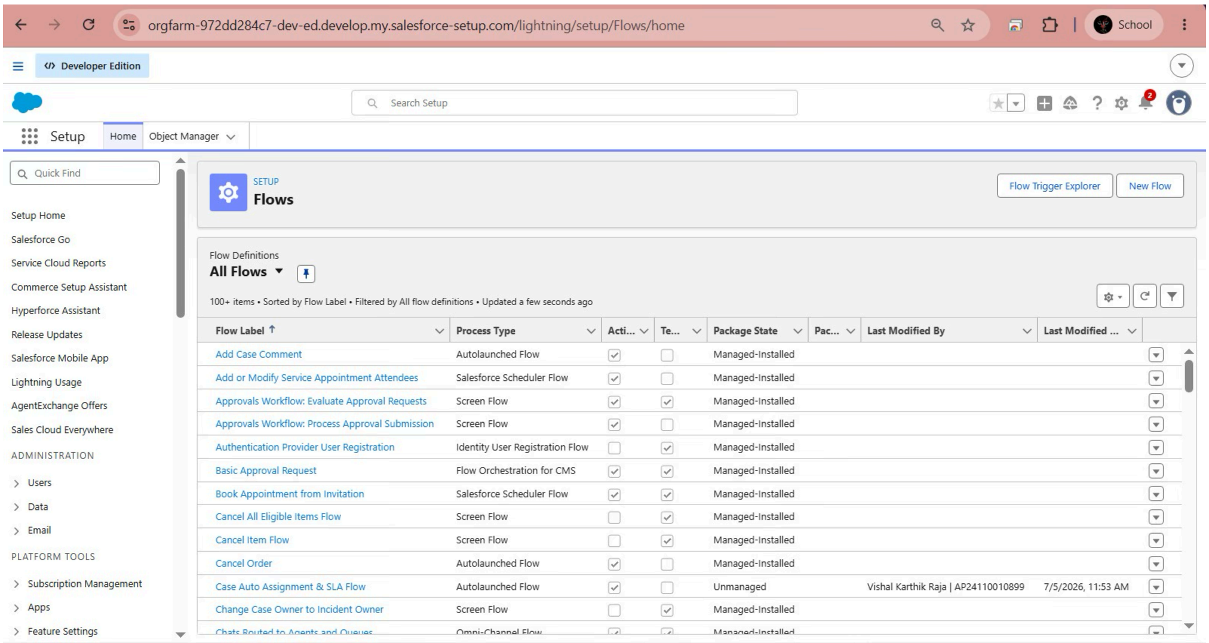
Salesforce Record-Triggered Flows were implemented to automate customer support operations.

Case Assignment Flow

This flow executes automatically whenever a Use Case record is created or updated. It evaluates the Issue Category and performs decision-based processing according to predefined business logic.

Case Auto Assignment & SLA Flow

This flow executes when a new Use Case is created. If the Priority is marked as High, the flow automatically assigns ownership of the case to the Damage Support Queue, reducing manual assignment efforts and improving response time.



5.7 Queue Implementation

Salesforce Queues were configured to distribute customer support requests among support teams.

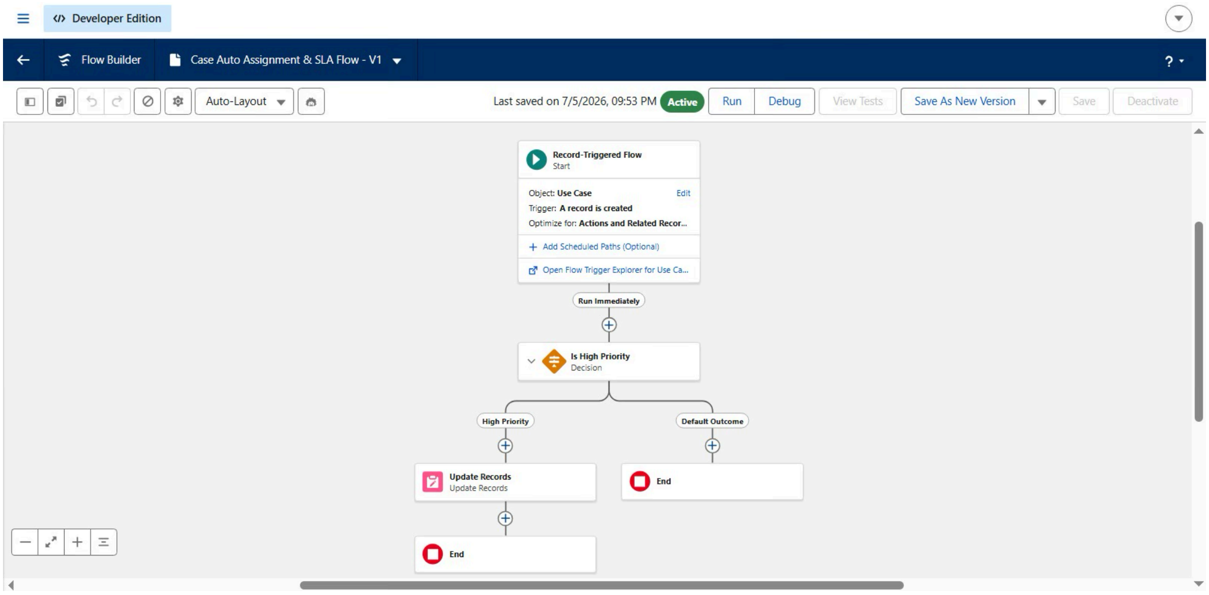
Damage Support Queue

Handles high-priority customer support requests related to damaged products.

Warranty Support Queue

Handles customer requests related to warranty claims and warranty services.

Queue implementation improves workload distribution and reduces manual case assignment.



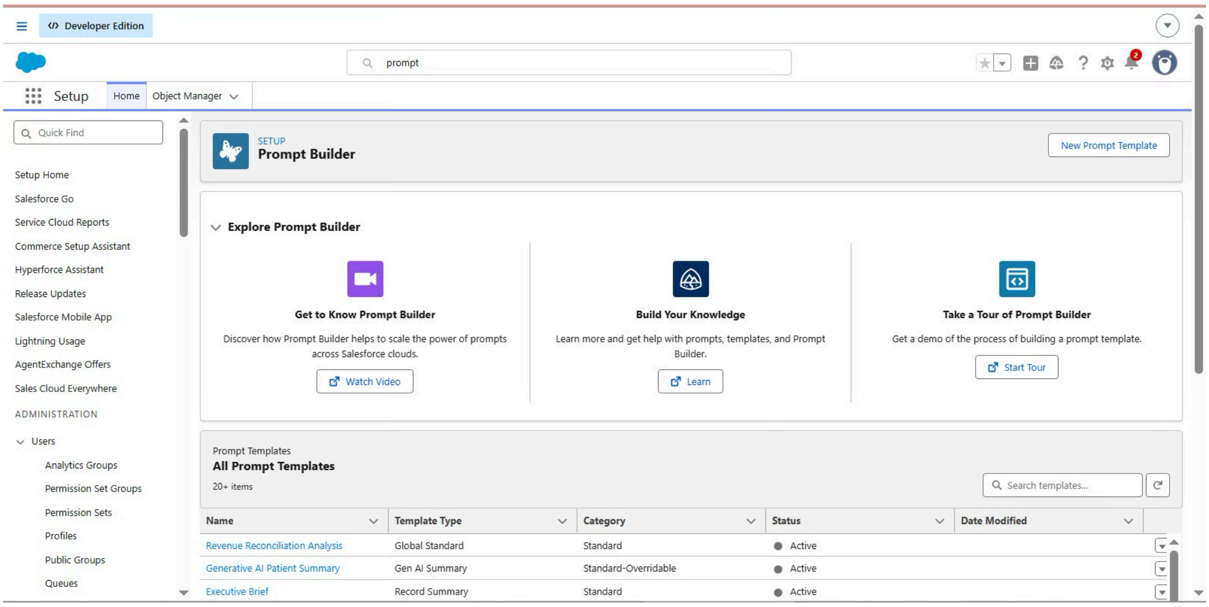
5.8 Prompt Builder Implementation

Salesforce Prompt Builder was configured to generate AI-powered customer support summaries.

Configuration Details:

Parameter	Value
Prompt Template	MetaDataInspection
Prompt Type	Field Generation
Object	Use Case
Output Field	AI Summary

The prompt template generates concise and professional summaries based on the available customer support information, improving documentation quality and assisting support personnel.



5.9 Lightning Application Implementation

A custom Lightning Application named AI-Powered Customer Support & Service Automation was developed using Salesforce Lightning App Builder.

The application provides centralized access to:

- Customer
- Product
- Order
- Use Case
- Reports
- Dashboards

The user interface was designed to simplify navigation while providing quick access to frequently used business objects.

5.10 Security Implementation

Security mechanisms were configured using Salesforce security features.

Implemented security controls include:

- Profiles
- Queue Ownership
- Field-Level Security
- Validation Rules
- Lightning Page Permissions

These controls ensure that only authorized users can access or modify application data.

5.11 AI Integration

Artificial Intelligence was integrated into the application using Salesforce Prompt Builder.

The AI component automatically generates professional summaries for customer support cases, reducing manual documentation efforts and ensuring consistent communication across support teams.

The AI Summary field stores generated content directly within the Use Case record, making it available for future reference.

5.12 Implementation Summary

The implementation phase successfully transformed the project requirements into a fully functional Salesforce application. Through the use of custom objects, relationships, validation rules, automation flows, queues, Lightning components, and AI-powered Prompt Builder, the system automates customer support processes while improving efficiency, data accuracy, and user productivity.

The completed application demonstrates the practical implementation of Salesforce CRM capabilities combined with Artificial Intelligence to build a scalable customer support solution.